L03 Draw a resistor

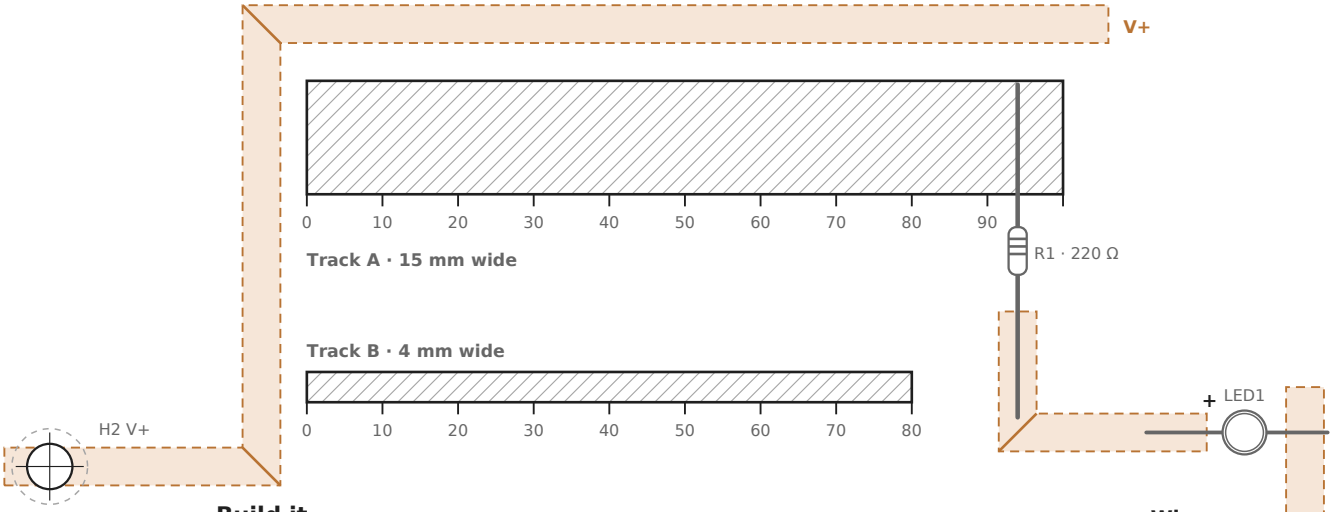
A pencil line conducts, a little. Slide a wiper along one to dim an LED.

Draft: tagged steps may change

H1 BUS A · NC

You need

- LED1: bright red LED, 5 mm, clear lens
- R1: 220 Ω (red-red-brown) **[P]**
- A soft 6B pencil and an eraser
- 5 mm copper tape and clear tape
- Scissors
- Power page P1; a multimeter

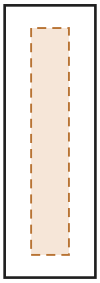


Build it

1. Shade track A and track B with the 6B pencil. Press hard and go over each one at least 10 times, until it shines. Stay inside the lines.
2. Rub out any pencil in the gap between track A and the V+ lane. A smudge there joins them.
3. Lay the three lanes, one piece of tape each. Fold at the creases; never cut.
4. Punch the holes and set eyelets in H2 and H3. **[T0]**
5. Lay R1's top lead flat across the end of track A and hold it down with clear tape. Its bottom lead goes on the lane below. Either way round works. **[T1]**
6. Place LED1 as printed: long leg on the +.
7. Hold R1's bottom lead and both LED1 leads flat with a patch of copper tape. **[T0]**
8. Cut out the wiper. Stick 30 mm of tape on it where marked, then turn it copper side down.
9. Bind this page with P1 and switch BT1 on. Press the wiper across the V+ lane and track A, then slide it.

[T0], **[T1]** and **[P]** mark choices that may change: **[T0]** after the contact tests, **[T1]** after the pencil tests, **[P]** once the power supply is settled.

Wiper

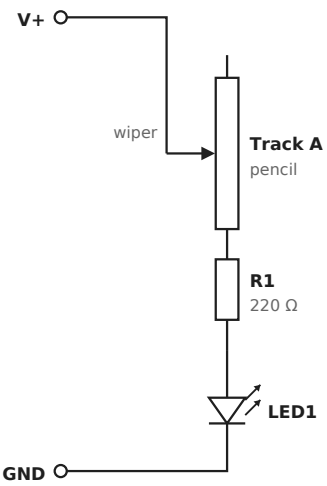


cut out

H3 GND

H4 BUS B · NC

Schematic



The wiper sets how much line is in the loop.

Predict Where will LED1 be brightest?

Wiper near R1 / far from R1 / no difference

Circle one. Why do you think so?

Measure Switch BT1 off. Probe each track at 0 and at a mark.

	10 mm	20 mm	40 mm	80 mm
Track A	kΩ	kΩ	kΩ	kΩ
Track B	kΩ	kΩ	kΩ	kΩ

Explain What does a longer line do to resistance?

And a narrower one? Use your Measure readings.

Print at 100 %: this bar should measure 100 mm.